

Class Test Chapter-1 relations and Functions 20 marks Class 12

1.	A function $f: [-4, 4] \rightarrow [0, 4]$ is given by $f(x) = \sqrt{16 - x^2}$. Show that f is an onto function but not a one-one function. Further, find all possible values of 'a' for which $f(a) = \sqrt{7}$.	5
2.	Let R be the relation defined in the set $A = \{1, 2, 3, 4, 5, 6, 7\}$ by $R = \{(a, b) : \text{both } a \text{ and } b \text{ are either odd or even}\}$. Show that R is an equivalence relation. Hence, find the elements of equivalence class $[1]$.	5
3.	Check whether the relation S in the set of real numbers R defined by $S = \{(a, b) : \text{where } a - b + \sqrt{2} \text{ is an irrational number}\}$ is reflexive, symmetric or transitive.	5
4.	If N denotes the set of all natural numbers and R is the relation on $N \times N$ defined by $(a, b) R (c, d)$, if $ad(b + c) = bc(a + d)$. Show that R is an equivalence relation.	5

Class Test Chapter-1 relations and Functions 20 marks Class 12

1.	A function $f: [-4, 4] \rightarrow [0, 4]$ is given by $f(x) = \sqrt{16 - x^2}$. Show that f is an onto function but not a one-one function. Further, find all possible values of 'a' for which $f(a) = \sqrt{7}$.	5
2.	Let R be the relation defined in the set $A = \{1, 2, 3, 4, 5, 6, 7\}$ by $R = \{(a, b) : \text{both } a \text{ and } b \text{ are either odd or even}\}$. Show that R is an equivalence relation. Hence, find the elements of equivalence class $[1]$.	5
3.	Check whether the relation S in the set of real numbers R defined by $S = \{(a, b) : \text{where } a - b + \sqrt{2} \text{ is an irrational number}\}$ is reflexive, symmetric or transitive.	5
4.	If N denotes the set of all natural numbers and R is the relation on $N \times N$ defined by $(a, b) R (c, d)$, if $ad(b + c) = bc(a + d)$. Show that R is an equivalence relation.	5

Class Test Chapter-1 relations and Functions 20 marks Class 12

1.	A function $f: [-4, 4] \rightarrow [0, 4]$ is given by $f(x) = \sqrt{16 - x^2}$. Show that f is an onto function but not a one-one function. Further, find all possible values of 'a' for which $f(a) = \sqrt{7}$.	5
2.	Let R be the relation defined in the set $A = \{1, 2, 3, 4, 5, 6, 7\}$ by $R = \{(a, b) : \text{both } a \text{ and } b \text{ are either odd or even}\}$. Show that R is an equivalence relation. Hence, find the elements of equivalence class $[1]$.	5
3.	Check whether the relation S in the set of real numbers R defined by $S = \{(a, b) : \text{where } a - b + \sqrt{2} \text{ is an irrational number}\}$ is reflexive, symmetric or transitive.	5
4.	If N denotes the set of all natural numbers and R is the relation on $N \times N$ defined by $(a, b) R (c, d)$, if $ad(b + c) = bc(a + d)$. Show that R is an equivalence relation.	5

Class Test Chapter-2 inverse t functions 20 marks Class 12

1.	Evaluate : $\sin^{-1}\left(\sin \frac{3\pi}{4}\right) + \cos^{-1}\left(\cos \frac{3\pi}{4}\right) + \tan^{-1}(1)$	3
2.	Draw the graph of $f(x) = \sin^{-1} x$, $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$. Also, write range of $f(x)$.	3
3.	Find the domain of $y = \sin^{-1}(x^2 - 4)$.	3
4.	Find the value of $\cos^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \operatorname{cosec}^{-1}(-2)$.	3
5.	Simplify : $\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right)$	5
6.	Find value of k if $\sin^{-1}\left[k \tan\left(2 \cos^{-1} \frac{\sqrt{3}}{2}\right)\right] = \frac{\pi}{3}$.	3

Class Test Chapter-2 inverse t functions 20 marks Class 12

1.	Evaluate : $\sin^{-1}\left(\sin \frac{3\pi}{4}\right) + \cos^{-1}\left(\cos \frac{3\pi}{4}\right) + \tan^{-1}(1)$	3
2.	Draw the graph of $f(x) = \sin^{-1} x$, $x \in \left[-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right]$. Also, write range of $f(x)$.	3
3.	Find the domain of $y = \sin^{-1}(x^2 - 4)$.	3
4.	Find the value of $\cos^{-1}\left(\frac{1}{2}\right) - \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) + \operatorname{cosec}^{-1}(-2)$.	3
5.	Simplify : $\tan^{-1}\left(\frac{\cos x}{1 - \sin x}\right)$	5
6.	Find value of k if $\sin^{-1}\left[k \tan\left(2 \cos^{-1} \frac{\sqrt{3}}{2}\right)\right] = \frac{\pi}{3}$.	3

Class Test Chapter-3 matrices 20 marks Class 12

1.	If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that $A^3 - 6A^2 + 7A + 2I = O$.	5
2.	Given a square matrix A of order 3 such that $A^2 = \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix}$, show that $A^3 = A^{-1}$.	5
3.	If $A = \begin{bmatrix} -1 & a & 2 \\ 1 & 2 & x \\ 3 & 1 & 1 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ -8 & 7 & -5 \\ b & y & 3 \end{bmatrix}$, find the value of $(a + x) - (b + y)$.	5
4.	If $f(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then prove that $f(\alpha) \cdot f(-\beta) = f(\alpha - \beta)$	5

Class Test Chapter-3 matrices 20 marks Class 12

1.	If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that $A^3 - 6A^2 + 7A + 2I = O$.	5
2.	Given a square matrix A of order 3 such that $A^2 = \begin{bmatrix} 3 & -4 & 4 \\ 0 & -1 & 0 \\ -2 & 2 & -3 \end{bmatrix}$, show that $A^3 = A^{-1}$.	5
3.	If $A = \begin{bmatrix} -1 & a & 2 \\ 1 & 2 & x \\ 3 & 1 & 1 \end{bmatrix}$ and $A^{-1} = \begin{bmatrix} 1 & -1 & 1 \\ -8 & 7 & -5 \\ b & y & 3 \end{bmatrix}$, find the value of $(a + x) - (b + y)$.	5
4.	If $f(\alpha) = \begin{bmatrix} \cos \alpha & -\sin \alpha & 0 \\ \sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then prove that $f(\alpha) \cdot f(-\beta) = f(\alpha - \beta)$	5

Class Test Chapter-4 determinants 20 marks Class 12

1.	(a) If $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$, find $(AB)^{-1}$.	5
2.	If $A = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, then find AB and use it to solve the following system of equations : $x - 2y = 3$ $2x - y - z = 2$ $-2y + z = 3$	5
3.	If matrix $A = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 1 & 3 \\ 1 & 1 & 1 \end{bmatrix}$, find A^{-1} and hence solve the following system of linear equations : $3x + 2y + z = 2000$ $4x + y + 3z = 2500$ $x + y + z = 900$	5
4.	Using determinants, find the area of ΔPQR with vertices $P(3, 1)$, $Q(9, 3)$ and $R(5, 7)$. Also, find the equation of line PQ using determinants.	5

Class Test Chapter-4 determinants 20 marks Class 12

1.	(a) If $A = \begin{bmatrix} 1 & 2 & -2 \\ -1 & 3 & 0 \\ 0 & -2 & 1 \end{bmatrix}$ and $B^{-1} = \begin{bmatrix} 3 & -1 & 1 \\ -15 & 6 & -5 \\ 5 & -2 & 2 \end{bmatrix}$, find $(AB)^{-1}$.	5
2.	If $A = \begin{bmatrix} -3 & -2 & -4 \\ 2 & 1 & 2 \\ 2 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 0 \\ -2 & -1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, then find AB and use it to solve the following system of equations : $x - 2y = 3$ $2x - y - z = 2$ $-2y + z = 3$	5
3.	If matrix $A = \begin{bmatrix} 3 & 2 & 1 \\ 4 & 1 & 3 \\ 1 & 1 & 1 \end{bmatrix}$, find A^{-1} and hence solve the following system of linear equations : $3x + 2y + z = 2000$ $4x + y + 3z = 2500$ $x + y + z = 900$	5
4.	Using determinants, find the area of ΔPQR with vertices $P(3, 1)$, $Q(9, 3)$ and $R(5, 7)$. Also, find the equation of line PQ using determinants.	5

Class Test Chapter-5 continuity and differentiability 20 marks Class 12

1.	If $y = (x + \sqrt{x^2 - 1})^2$, then show that $(x^2 - 1) \left(\frac{dy}{dx} \right)^2 = 4y^2$.	3
2.	If $y = \sqrt{ax + b}$, prove that $y \left(\frac{d^2y}{dx^2} \right) + \left(\frac{dy}{dx} \right)^2 = 0$.	3
3.	If $x = a \sin 2t$, $y = a(\cos 2t + \log \tan t)$, then find $\frac{dy}{dx}$.	3
4.	Find the value(s) of ' λ ', if the function $f(x) = \begin{cases} \frac{\sin^2 \lambda x}{x^2}, & \text{if } x \neq 0 \text{ is continuous at } x = 0. \\ 1, & \text{if } x = 0 \end{cases}$	3
5.	If $y = \tan x + \sec x$, then prove that $\frac{d^2y}{dx^2} = \frac{\cos x}{(1 - \sin x)^2}$.	3
6.	If $x \cos (p + y) + \cos p \sin (p + y) = 0$, prove that $\cos p \frac{dy}{dx} = -\cos^2 (p + y)$, where p is a constant.	5

Class Test Chapter-5 continuity and differentiability 20 marks Class 12

1.	If $y = (x + \sqrt{x^2 - 1})^2$, then show that $(x^2 - 1) \left(\frac{dy}{dx} \right)^2 = 4y^2$.	3
2.	If $y = \sqrt{ax + b}$, prove that $y \left(\frac{d^2y}{dx^2} \right) + \left(\frac{dy}{dx} \right)^2 = 0$.	3
3.	If $x = a \sin 2t$, $y = a(\cos 2t + \log \tan t)$, then find $\frac{dy}{dx}$.	3
4.	Find the value(s) of ' λ ', if the function $f(x) = \begin{cases} \frac{\sin^2 \lambda x}{x^2}, & \text{if } x \neq 0 \text{ is continuous at } x = 0. \\ 1, & \text{if } x = 0 \end{cases}$	3
5.	If $y = \tan x + \sec x$, then prove that $\frac{d^2y}{dx^2} = \frac{\cos x}{(1 - \sin x)^2}$.	3
6.	If $x \cos (p + y) + \cos p \sin (p + y) = 0$, prove that $\cos p \frac{dy}{dx} = -\cos^2 (p + y)$, where p is a constant.	5

Class Test Chapter-6 application of derivatives 20 marks Class 12

1.	The median of an equilateral triangle is increasing at the rate of $2\sqrt{3}$ cm/s. Find the rate at which its side is increasing.	3
2.	Sum of two numbers is 5. If the sum of the cubes of these numbers is least, then find the sum of the squares of these numbers.	3
3.	Given that $f(x) = \frac{\log x}{x}$, find the point of local maximum of $f(x)$.	3
4.	Find the intervals in which the function $f(x) = x^4 - 4x^3 + 4x^2 + 15$, is strictly increasing.	3
5.	The volume of a cube is increasing at the rate of $6 \text{ cm}^3/\text{s}$. How fast is the surface area of cube increasing, when the length of an edge is 8 cm ?	3
6.	The perimeter of a rectangular metallic sheet is 300 cm. It is rolled along one of its sides to form a cylinder. Find the dimensions of the rectangular sheet so that volume of cylinder so formed is maximum.	5

Class Test Chapter-6 application of derivatives 20 marks Class 12

1.	The median of an equilateral triangle is increasing at the rate of $2\sqrt{3}$ cm/s. Find the rate at which its side is increasing.	3
2.	Sum of two numbers is 5. If the sum of the cubes of these numbers is least, then find the sum of the squares of these numbers.	3
3.	Given that $f(x) = \frac{\log x}{x}$, find the point of local maximum of $f(x)$.	3
4.	Find the intervals in which the function $f(x) = x^4 - 4x^3 + 4x^2 + 15$, is strictly increasing.	3
5.	The volume of a cube is increasing at the rate of $6 \text{ cm}^3/\text{s}$. How fast is the surface area of cube increasing, when the length of an edge is 8 cm ?	3
6.	The perimeter of a rectangular metallic sheet is 300 cm. It is rolled along one of its sides to form a cylinder. Find the dimensions of the rectangular sheet so that volume of cylinder so formed is maximum.	5

Class Test Chapter-8 differential Equations 20 marks Class 12

1.	Find the particular solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}, y(1) = 0.$	5
2.	Find the particular solution of the differential equation $\frac{dy}{dx} - 2xy = 3x^2 e^{x^2}; y(0) = 5.$	5
3.	Find the particular solution of the differential equation $(xe^{\frac{y}{x}} + y) dx = x dy$, given that $y = 1$ when $x = 1$.	5
4.	Find the particular solution of the differential equation $\left[x \sin^2\left(\frac{y}{x}\right) - y \right] dx + x dy = 0$ given that $y = \frac{\pi}{4}$, when $x = 1$.	5

Class Test Chapter-8 differential Equations 20 marks Class 12

1.	Find the particular solution of the differential equation $\frac{dy}{dx} = \frac{x+y}{x}, y(1) = 0.$	5
2.	Find the particular solution of the differential equation $\frac{dy}{dx} - 2xy = 3x^2 e^{x^2}; y(0) = 5.$	5
3.	Find the particular solution of the differential equation $(xe^{\frac{y}{x}} + y) dx = x dy$, given that $y = 1$ when $x = 1$.	5
4.	Find the particular solution of the differential equation $\left[x \sin^2\left(\frac{y}{x}\right) - y \right] dx + x dy = 0$ given that $y = \frac{\pi}{4}$, when $x = 1$.	5